

Hospital HMIS Data Analysis

Interactive Healthcare Analytics Dashboard — Power BI & SQL

PATIENT & ADMISSION

HOSPITAL OPERATIONS

BILLING & PAYMENT

DRUG & TREATMENT

INSURANCE



Project Overview

Turning hospital operational data into interactive, decision-support dashboards.

Objective

Power BI dashboard exploring patients, admissions, billing, drugs, treatment, and insurance.

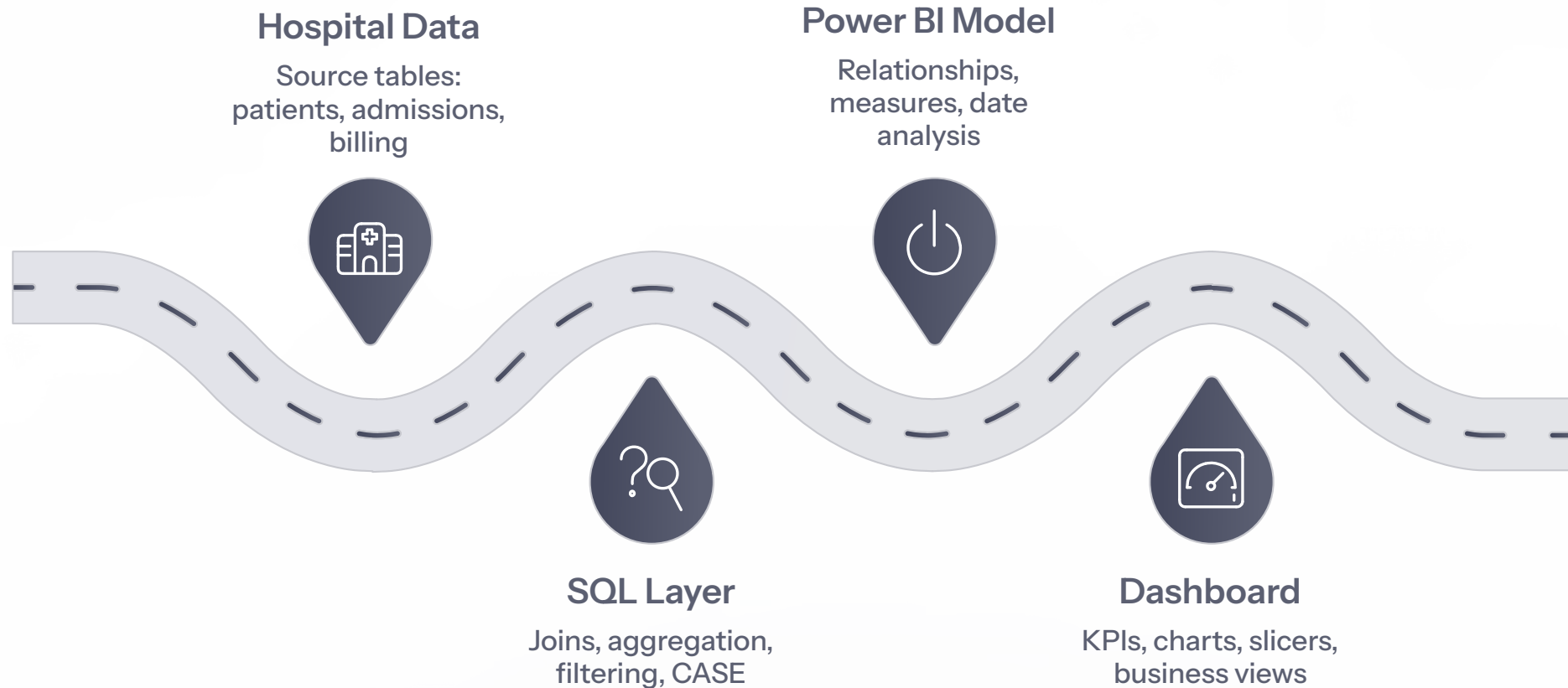
Tools

Power BI · SQL/MySQL · DAX · Data Modeling · Interactive Filters & KPIs

Outcome

Multi-page analytical report for monitoring and interpreting healthcare data.

Data & Analytical Workflow



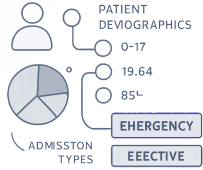
The SQL layer supports structured analytical questions — department treatment cost, disease treatment cost, and patient/admission patterns.

Dashboard Pages



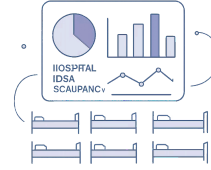
1. Healthcare Overview

High-level hospital KPIs and trend-oriented overview



2. Patient & Admission

Patients, age groups, admission types, departments, doctors, length of stay



3. Hospital Operations

Beds, occupancy, wards, departments, operational indicators



4. Billing & Payment

Treatment cost, patient payable, payment status, billing trends



5. Drug & Treatment

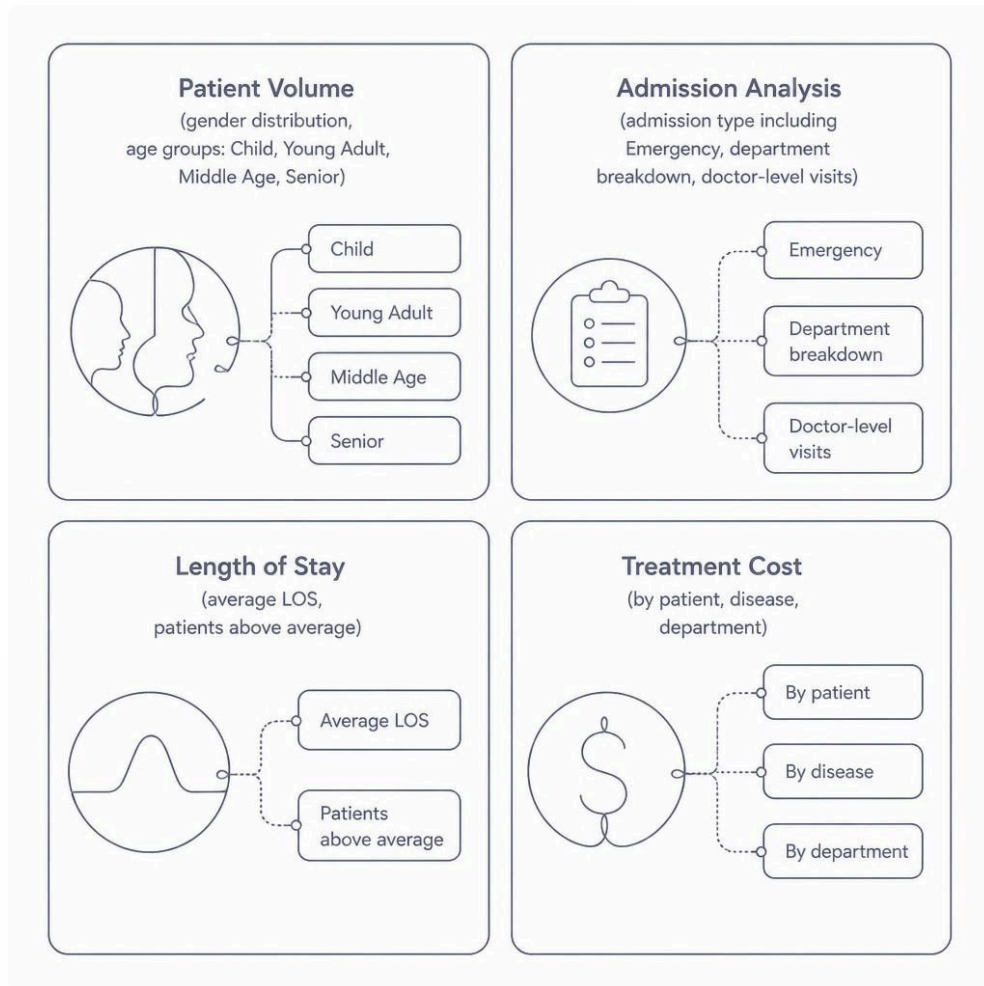
Drug usage, cost, categories, diseases, treatment analysis



6. Insurance Analysis

Coverage limits, coverage %, provider analysis, disease/department coverage

Patient & Admission Analysis



SQL Techniques

COUNT, GROUP BY, WHERE, JOIN, and CASE for patient and admission analysis.

Cost Analysis

Treatment costs examined by patient, disease, and department.

Advanced Analysis

Subqueries identify patients above average treatment cost and length of stay.

Hospital Operations

Bed Indicators

Total, available, and occupied bed counts with status distribution

Ward & Department Views

Operational breakdown by ward type and department

Length of Stay

Average LOS by department with year/date-based trend analysis

Interactive Slicers

Time and category filters for focused operational exploration



Billing, Drug & Treatment Analysis

Billing & Payment

- Total treatment cost & patient payable
- Payment status & billing trends
- Cost by department

Drug & Treatment

- Total drugs used & drug cost
- Drug category analysis
- Usage by disease/department

SQL Techniques

Aggregations

Multi-table JOINS

AVG / MIN / MAX

HAVING

Subqueries

DENSE_RANK()

Insurance Analysis



Key Measures

Business Questions

Which provider types show different coverage patterns? Which diseases/departments account for more insurance-covered amounts?

→ Coverage limit & coverage %

→ Patient payable amount

→ Insurance covered amount

→ Payment status by provider

Key SQL Techniques Demonstrated



JOINS

Connect patient, admission, department, disease, and billing tables.



GROUP BY & HAVING

Summarize patients and costs; filter departments above thresholds.



CASE Expressions

Create age groups from patient age values.



Subqueries

Compare individual values against overall averages.



Window Functions

DENSE_RANK() ranks departments by total treatment cost.

Project Takeaways

1 Multi-page Power BI dashboard

Operational, financial, treatment, and insurance perspectives

2 Structured SQL analysis

Aggregation, joins, filtering, subqueries, CASE, and ranking across hospital tables

3 Interactive reporting experience

Exploratory healthcare analytics for decision-makers

